

Aitian Ma

Ph.D. Candidate in Computer Science | Machine Learning, LLM Reasoning and Search, and Trustworthy AI
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Research Interests

Integrating LLM reasoning, search, and optimization with machine learning; LLM-based diagnosis and decision support; time-series modeling and time-series foundation models; robustness under distribution shift; trustworthy, privacy-preserving, and efficient AI; applications to health and medicine, sensing, and networked systems.

Education

Florida International University, Miami, FL Jan 2022 – Dec 2026 (expected)

Ph.D. in Computer Science, Knight Foundation School of Computing and Information Sciences
Dissertation: *Creating Reliable and Adaptive Wireless Sensing Systems via Resource-Efficient Learning*
Advisor: Mo Sha | AI research co-advisor: Dongsheng Luo

Beijing University of Technology, Beijing, China 2010 – 2013

M.S. in Computer Science

Henan Agricultural University, Zhengzhou, China 2006 – 2010

B.S. in Computer Science

Honors and Awards

- Best Paper Award, AI4TS Workshop on AI for Time Series Analysis, 2023.
- Best Paper Award Nominee, IEEE DCOSS-IoT, 2025.
- 3rd Place, Microsoft HPC Campus Programming Competition.

Research Experience

Graduate Research Assistant, Florida International University Jan 2022 – Present

- **LLM reasoning for diagnosis and optimization.** Co-developed LORADOCTOR (ICMLA 2025, co-first author) with University College London: an LLM-driven framework that combines causal analysis of heterogeneous measurements, constraint-aware policy optimization, and performance prediction with natural-language explanations, evaluated on a year-long deployment with more than 840,000 records. Developing dual-mind LLM methods with Yale University that pair fast pattern-based responses with deliberate multi-step planning.
- **Search control for LLM agents.** Developed LeanRepair, a deterministic external controller for multi-candidate LLM search (compared with Self-Refine, Tree-of-Thoughts, and Graph-of-Thoughts) that detects oscillating and stagnating trajectories, prevents recurrence of failed candidates, and reduces token use by a median 23.9% while raising pass rates, across four open LLM families from 0.6B to 14B parameters (under review, ICLR 2027).
- **Time-series machine learning.** Developed MixLinear (ICLR 2026), a 0.1K-parameter multivariate forecaster, and MMFNet (ACM SAC 2026); co-authored Parametric Augmentation for Time Series Contrastive Learning (ICLR 2024); developed SpecSpin (4-bit quantization) and RosKV (KV-cache compression) for time-series foundation models (both under review, ICLR 2027), evaluated on NVIDIA A100 GPUs.
- **Simulation-to-reality adaptation.** Designed contrastive and adversarial domain-adaptation methods for industrial wireless mesh network configuration (WMN-CDA, ACM SAC 2025; IEEE/ACM ToN 2024), validated across multiple simulators and a physical WirelessHART testbed.
- **Sensorless and energy-harvesting sensing.** Built LoRa-based environmental sensing systems that infer air temperature from radio link characteristics (DCOSS-IoT 2025, Best Paper Nominee) and operate with energy harvesting and domain adaptation (ICCCN 2024).
- **Grant development.** Assisted PI Mo Sha with 2025 proposals on a multi-agent digital-twin framework for multi-hazard bridge resilience (IBT/ABC-UTC) and on securing large language models against jailbreaking attacks

(Google Research), including technical approach, literature review, and broader impacts.

Data Engineer and Instructor, Chinese Academy of Sciences, Beijing

Jul 2013 – Dec 2021

- Managed large-scale text and business-data platforms (Elasticsearch, Logstash, Kibana) supporting search, credit analytics, sentiment analysis, financial event detection, anomaly detection, and alert generation.
- Maintained 2,000+ Scrapy-based data-extraction templates and built a Neo4j knowledge graph with millions of nodes for advanced search and knowledge discovery.
- Administered Linux-based data and web platforms and distributed data workflows for research users.

Software Engineer Intern, Ericsson China R&D Center, Beijing

Jun 2012 – Sep 2012

- Built trouble-report analytics and dashboards for LTE, microwave, and O&M teams using Perl, Shell, Java, and PHP.

Publications

“*” denotes equal contribution.

Journal Articles

- [J1] Junyang Shi, **Aitian Ma**, Xia Cheng, Mo Sha, and Xi Peng. “Adapting Wireless Network Configuration from Simulation to Reality via Deep Learning Based Domain Adaptation.” *IEEE/ACM Transactions on Networking*, 32(3):1983–1998, 2024.

Conference and Workshop Papers

- [C1] **Aitian Ma**, Dongsheng Luo, and Mo Sha. “MixLinear: Extreme Low-Resource Multivariate Time Series Forecasting with 0.1K Parameters.” *International Conference on Learning Representations (ICLR)*, 2026.
- [C2] **Aitian Ma**, Dongsheng Luo, and Mo Sha. “MMFNet: Multi-Scale Frequency Masking Neural Network for Multivariate Time Series Forecasting.” *ACM/SIGAPP Symposium on Applied Computing (SAC)*, 2026. Acceptance rate 25%.
- [C3] Dongyi Ma*, **Aitian Ma***, Sirui Luo, Martin de Jode, Andrew Hudson-Smith, and Mo Sha. “LLM-Driven Diagnosis and Adaptive Policy Optimization for Reducing Packet Error Rate in LoRaWAN Networks.” *IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2025. Acceptance rate 16.6%.
- [C4] **Aitian Ma**, Jean Tondoy Rodriguez, Mo Sha, and Dongsheng Luo. “Sensorless Air Temperature Sensing Using LoRa Link Characteristics.” *IEEE DCOSS-IoT*, 2025. Acceptance rate 23.6%. **Best Paper Award Nominee.**
- [C5] **Aitian Ma**, Jean Tondoy Rodriguez, and Mo Sha. “A LoRa-Based Energy-Harvesting Sensing System for Living Environment.” *IEEE International Workshop on Metrology for Living Environment (MetroLivEnv)*, 2025.
- [C6] **Aitian Ma** and Mo Sha. “WMN-CDA: Contrastive Domain Adaptation for Wireless Mesh Network Configuration.” *ACM SAC, Cyber-Physical Systems Track*, 2025. Acceptance rate 23.8%.
- [C7] **Aitian Ma**, Jean Tondoy Rodriguez, and Mo Sha. “Enabling Reliable Environmental Sensing with LoRa, Energy Harvesting, and Domain Adaptation.” *IEEE International Conference on Computer Communications and Networks (ICCCN)*, 2024. Acceptance rate 29.9%.
- [C8] Xu Zheng, Tianchun Wang, Wei Cheng, **Aitian Ma**, Haifeng Chen, Mo Sha, and Dongsheng Luo. “Parametric Augmentation for Time Series Contrastive Learning.” *International Conference on Learning Representations (ICLR)*, 2024. Acceptance rate 31.1%.
- [C9] Xu Zheng, Tianchun Wang, Wei Cheng, **Aitian Ma**, Haifeng Chen, Mo Sha, and Dongsheng Luo. “AutoTCL: Automated Time Series Contrastive Learning with Adaptive Augmentations.” *AI4TS Workshop*, 2023. **Best Paper Award.**

Manuscripts Under Review

- [U1] “SpecSpin: Spectral Truncation and Rotation for 4-Bit Zero-Shot Time-Series Foundation Models.” Under double-blind review, ICLR 2027.
- [U2] “RosKV: Robust Spectral Key-Value Cache Compression for Time-Series Foundation Models.” Under double-blind review, ICLR 2027.
- [U3] “LeanRepair: Token-Efficient External Control for Local Code Repair.” Under double-blind review, ICLR 2027.
- [U4] **Aitian Ma**, Jean Tondoy Rodriguez, Mo Sha, and Dongsheng Luo. “Sensorless Air Temperature Sensing Using LoRa Link Characteristics.” Journal extension, major revision, *ACM Transactions on Sensor Networks*.
- [U5] **Aitian Ma**, Dongsheng Luo, Ali Maatouk, Rex Ying, and Mo Sha. “Zero-Shot Industrial Wireless Mesh Network Configuration via Dual-Mind LLMs.” Under review.

- [U6] **Aitian Ma**, Ali Maatouk, Rex Ying, and Mo Sha. “NoiseFlow: Noise-Enhanced Flow Matching for Domain Adaptation in Industrial Wireless Mesh Network Configuration.” Under review.
- [U7] **Aitian Ma** and Mo Sha. “LLM-Based Noise-Aware Domain Adaptation Agent for Industrial Wireless Mesh Network Configuration.” Under review.
- [U8] **Aitian Ma** and Mo Sha. “Large Language Model Based Data Synthesis for Industrial Wireless Network Configuration.” Under review.

Teaching Experience

Teaching Assistant, Florida International University

2022 – 2025

- COP 4338 Programming III (Java, object-oriented design, collections, multithreading, networking, unit testing).
- COP 5522 Parallel and Distributed Computing (graduate; MPI, CUDA/GPU programming, MapReduce, cloud models).
- COP 4710 Database Management; CEN 4010 Software Engineering; COP 4610 Operating Systems.
- Led lab sessions, held office hours, graded programming projects, and helped students debug systems-level code.

Instructor, Chinese Academy of Sciences, Beijing

2015 – 2019

- Taught Big Data Platform and Computer Networking workshops using demonstrations, hands-on exercises, and projects; engaged more than 2,000 students across educational levels and mentored more than 50 technical interns.

Mentoring

- Mentored 10+ undergraduate researchers at FIU in machine learning, wireless sensing, and systems projects; mentee Jean Tondoy Rodriguez co-authored three published papers and a journal manuscript, including the DCOSS-IoT 2025 Best Paper Nominee.

Research Collaborations

Yale University (Rex Ying, Ali Maatouk) — LLM-based network configuration; University College London, Centre for Advanced Spatial Analysis (Andrew Hudson-Smith) — LoRaWAN diagnosis; NEC Laboratories America (Wei Cheng, Haifeng Chen) — time-series representation learning.

Professional Service

Peer reviewer for IEEE and ACM journals and conferences.

Technical Skills

Languages: Python, C/C++, Java, SQL, Shell, JavaScript, Perl, PHP. **ML:** PyTorch, deep learning, LLM applications, time-series forecasting, contrastive learning, domain adaptation, anomaly detection. **Data and systems:** Linux, Elasticsearch/ELK, Neo4j, Scrapy, Django, Spring Boot, MPI, CUDA, LoRa/LoRaWAN, WirelessHART, NS-3, OMNeT++.

References

Deng Pan (Associate Professor, FIU, pan@fiu.edu) | Xuyu Wang (Assistant Professor, FIU, xuywang@fiu.edu) | Yanzhao Wu (Assistant Professor, FIU, yawu@fiu.edu)